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ECEN 474/704
Lab 7: Prelab
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1. **Comparison of inverting-amplifier topologies.** Rank the four amplifiers in gain, input impedance, output impedance, and linearity. Give expressions for the first three quantities.

   Solution.

   The comparison assumes comparable bias currents and device parameters. The ratings favor high voltage gain, high input impedance, and low output impedance for driving a load.

Specification	Current mirror	Digital CMOS	PMOS buffer	CMOS buffer
Voltage gain	Medium	Good	Poor	Poor
Input impedance	Good	Good	Good	Good
Output impedance	Poor	Poor	Good	Good
Linearity	Medium	Poor	Good	Medium

   The buffers have a target gain of -1 and are linear in the ideal square-law model. Tracking between devices of the same type favors the PMOS pair, although body effect and other non-idealities limit its linearity.

   (a) Gain and output resistance. Let $g_{oi} = 1/r_{oi}$ and $g_o = g_{o1} + g_{o2}$. With the supplies and bias-reference node at AC ground,

Circuit	A_{v0}	R_{out}
Current-mirror load	$-g_{m1}/g_o$	$1/g_o$
Digital CMOS	$-(g_{m1} + g_{m2})/g_o$	$1/g_o$
Self-biased PMOS	$-g_{m1}/(g_{m2} + g_{mb2} + g_o)$	$1/(g_{m2} + g_{mb2} + g_o)$
Self-biased CMOS	$-g_{m1}/(g_{m2} + g_o)$	$1/(g_{m2} + g_o)$

   The output resistance is evaluated with the input held constant. A finite R_L adds $1/R_L$ to each denominator. The PMOS load's source is the output while its body is fixed, giving the g_{mb2} term. The CMOS load's source and body are both fixed at V_{DD}.

   (b) Input impedance. Ignoring gate leakage, all four circuits have $R_{\text{in,DC}} \to \infty$. At finite frequency, let $k(s) = v_o/v_A$, where v_A is the gate node after R_S. Then

$$Z_{\text{in},A}(s) = \frac{1}{s\{C_A + C_C[1 - k(s)]\}} \approx \frac{1}{s[C_A + C_C(1 + |A_{v0}|)]}.$$

   The source sees $R_S + Z_{\text{in},A}$. For the single-input-device circuits, $C_A \approx C_{gs1}$; digital CMOS has $C_A \approx C_{gs1} + C_{gs2}$. Similarly, $C_C = C_m + C_{gd1}$, with C_{gd2} added for digital CMOS. Thus the equal DC input-resistance ratings do not imply equal AC input loading.

2. **Miller-theorem derivation and comparison.** Derive the generic amplifier transfer function using Miller's theorem, compare it with the nodal result, and discuss the approximation's utility.

Solution.

(a) Miller equivalent. Let v_A be the input node after R_A, let v_o be the output, and define $A_0 = G_m R_B > 0$, so the DC voltage gain is $-A_0$. For the capacitor C_C between these nodes,

$$i_A = sC_C(v_A - v_o) = sC_C[1 - k(s)]v_A,$$
$$i_o = sC_C(v_o - v_A) = sC_C[1 - 1/k(s)]v_o, \qquad k(s) = \frac{v_o}{v_A}.$$

These identities are exact if the frequency-dependent $k(s)$ is retained. Replacing $k(s)$ by its low-frequency value $-A_0$ gives

$$\boxed{C_{\text{in}} = C_A + C_C(1 + A_0), \qquad C_{\text{out}} = C_B + C_C(1 + 1/A_0).}$$

The input divider and output transconductance stage then give

$$\frac{v_A}{v_{\text{in}}} \approx \frac{1}{1 + sR_A C_{\text{in}}}, \qquad \frac{v_o}{v_A} \approx \frac{-G_m R_B}{1 + sR_B C_{\text{out}}}.$$

Thus the constant-gain Miller approximation is

$$\boxed{H_M(s) \approx \frac{-A_0}{(1 + sR_A C_{\text{in}})(1 + sR_B C_{\text{out}})}.}$$

The resulting time constants are $R_A C_{\text{in}}$ and $R_B C_{\text{out}}$. Their accuracy as pole estimates depends on the frequency variation of the inter-node gain.

(b) Exact nodal result. With the controlled current $G_m v_A$ directed from the output to AC ground, the two node equations are

$$\left[\frac{1}{R_A} + s(C_A + C_C)\right] v_A - sC_C v_o = \frac{v_{\text{in}}}{R_A},$$
$$(G_m - sC_C)v_A + \left[\frac{1}{R_B} + s(C_B + C_C)\right] v_o = 0.$$

Eliminating v_A yields

$$\boxed{H(s) = \frac{-A_0(1 - sC_C/G_m)}{1 + as + bs^2},}$$

where

$$a = R_A(C_A + C_C) + R_B(C_B + C_C) + G_m R_A R_B C_C,$$
$$b = R_A R_B(C_A C_B + C_A C_C + C_B C_C).$$

The denominator and zero agree with the handout. The overall minus sign makes $H(0) = -G_m R_B$, consistent with an inverting amplifier. The exact inter-node gain is

$$k(s) = -\frac{G_m - sC_C}{1/R_B + s(C_B + C_C)}.$$

Using this $k(s)$ in the Miller identities recovers the same nodal result.

(c) Comparison and utility. Expanding H_M gives coefficients $a_M = R_A C_{\text{in}} + R_B C_{\text{out}}$ and $b_M = R_A R_B C_{\text{in}} C_{\text{out}}$. Their differences from the exact denominator are

$$a_M - a = \frac{R_B C_C}{A_0} = \frac{C_C}{G_m},$$
$$b_M - b = R_A R_B \left[\frac{C_A C_C}{A_0} + A_0 C_B C_C + C_C^2 \left(2 + A_0 + \frac{1}{A_0} \right) \right].$$

The constant-gain approximation also removes the right-half-plane zero

$$\boxed{s_z = +\frac{G_m}{C_C}, \qquad f_z = \frac{G_m}{2\pi C_C}.}$$

Miller's theorem itself is an exact circuit identity; the loss of accuracy comes from substituting the constant $-A_0$ for $k(s)$. The approximation therefore provides an estimate of input loading and an initial compensation capacitance. The second pole and phase near unity gain require the frequency-dependent gain or the nodal result.

For $A_0 \gg 1$, negligible C_A, and a dominant Miller time constant,

$$f_{p1} \approx \frac{1}{2\pi A_0 R_A C_C}, \qquad \text{GBW} \approx A_0 f_{p1} \approx \frac{1}{2\pi R_A C_C}.$$

For $R_A = 100\,\text{k}\Omega$ and GBW $= 2\,\text{MHz}$, this gives

$$C_C \approx 0.7958\,\text{pF}.$$

This is the total feedback capacitance, including C_{gd}; the external C_m is smaller by the relevant parasitic capacitance. The full two-node expression provides the corresponding refinement.

(d) Frequency and phase conventions. When $a^2 \geq 4b$, the two real poles are at $s = -\omega_{p1}$ and $s = -\omega_{p2}$, where

$$\omega_{p1} = \frac{2}{a + \sqrt{a^2 - 4b}}, \qquad \omega_{p2} = \frac{a + \sqrt{a^2 - 4b}}{2b}, \qquad f_{pi} = \frac{\omega_{pi}}{2\pi}.$$

Define the measured bandwidth f_{BW} by $|H(j2\pi f_{\text{BW}})| = A_0/\sqrt{2}$. It is approximately f_{p1} only when the other pole and the zero are sufficiently far away. The measured gain–bandwidth product is GBW $= A_0 f_{\text{BW}}$, while the unity-gain frequency satisfies $|H(j2\pi f_{\text{UGF}})| = 1$.

After removing the DC inversion, the dynamic phase lag is

$$\phi_{\text{lag}}(f) = \tan^{-1}\left(\frac{f}{f_{p1}}\right) + \tan^{-1}\left(\frac{f}{f_{p2}}\right) + \tan^{-1}\left(\frac{f}{f_z}\right).$$

The RHP zero adds phase lag. Therefore, $f_{p2} \geq \sqrt{3}\,\text{GBW}$ alone guarantees neither the exact UGF nor a lag below 120° when the zero is appreciable. For a monotone low-pass buffer with $A_0 = 1$, a positive-frequency unity-gain crossing need not exist. The buffer phase is therefore evaluated at f_{BW}, as specified in the handout.

3. **Design of the four inverting amplifiers.** Design the current-mirror-loaded inverter, digital CMOS inverter, self-biased PMOS inverter, and self-biased CMOS inverter in Figs. 7-5, 7-6, 7-8, and 7-9, respectively. All four use $V_{DD} = +0.9\,\text{V}$, $V_{SS} = -0.9\,\text{V}$, $R_S = 100\,\text{k}\Omega$, and GBW = 2 MHz.

Solution.

Common specifications and device assumptions. The high-gain designs require $A_0 = 10^{30/20} = 31.6228$, $C_L = 1\,\text{pF}$, and dynamic phase lag below 120° at UGF. The two diode-load designs require $A_0 = 1$ and $C_L = 25\,\text{pF}$. Here $A_0 = |A_{v0}|$ is positive; the voltage gain is negative. The nominal output is selected as $V_{o,Q} = 0$.

The device parameters are taken from Table I of the Lab 1 MOS characterization report, measured at single-finger $W/L = 5/0.27\,\mu\text{m}$:

Parameter	NFET	PFET
$K_P = \mu C_{ox}$ ($\mu\text{A/V}^2$)	215.6	47.57
$\lvert V_{TH}\rvert$ (V)	0.4330	0.4068
λ at $L_0 = 0.27\,\mu\text{m}$ (V^{-1})	0.1624	0.1945
θ (V^{-1})	0.9999	0.6558

The DC sizing uses the square-law model. Channel-length modulation is retained only through the small-signal estimate $g_o \approx \lambda I_D$:

$$I_D = \frac{1}{2} K_P \frac{W_{\text{eff}}}{L} V_{OV}^2, \qquad g_m = \frac{2I_D}{V_{OV}}, \qquad r_o \approx \frac{1}{\lambda I_D}.$$

The numerical designs initially omit body effect, mobility degradation, and DC channel-length-modulation corrections. The listed θ values are not applied to the square-law results. For lengths different from L_0, the assumed scaling is

$$\lambda_n(L) = 0.1624 \frac{L_0}{L}, \qquad \lambda_p(L) = 0.1945 \frac{L_0}{L},$$

while holding K_P and V_{TH} fixed. This length dependence is a design approximation that has not been verified by device characterization.

The frequency calculation assumes $C_A = 0$, $C_B = C_L$, and $C_C = C_m$. Gate, junction, and interconnect capacitances in the PDK modify this model. The results below are analytical estimates; their accuracy requires circuit simulation.

Width and multiplicity convention. Each table gives both total effective width and width per finger:

$$W_{\text{eff}} = w\, n_f\, m.$$

Parts (a)–(d) derive continuous w_{ana} and L_{ana}. The later grid-constrained table gives the initial dimensions, with w, L on the $0.01\,\mu\text{m}$ grid and initial $n_f = m = 2$; both counts remain adjustable. DC and AC verification apply to the selected geometry; any subsequent dimensional change requires recalculation and simulation.

(a) Current-mirror-loaded inverter (Fig. 7-5).

Transconductance and bias current. M1 is the NMOS driver; M2 and M3 form a 1:1 PMOS mirror. The handout's second-pole estimate would give

$$g_{m1} \gtrsim 2\pi(\sqrt{3})(2\,\text{MHz})(1\,\text{pF}) = 21.77\,\mu\text{S}.$$

This bound neglects the RHP zero. A larger value, $g_{m1} = 300\,\mu\text{S}$, is selected, with phase evaluated using the complete two-node transfer function. At $L = 0.270\,\mu\text{m}$, $\lambda_\Sigma = \lambda_n + \lambda_p = 0.3569\,\text{V}^{-1}$. The required current and overdrive are

$$I_D = I_{\text{bias}} = \frac{g_{m1}}{A_0\lambda_\Sigma} = \frac{300\times10^{-6}}{31.6228(0.3569)} = 26.581\,\mu\text{A},$$
$$R_{\text{out}} = \frac{1}{\lambda_\Sigma I_D} = 105.409\,\text{k}\Omega,$$
$$V_{OV1} = \frac{2I_D}{g_{m1}} = 0.177208\,\text{V}.$$

The choice $g_{m2} = g_{m3} = g_{m1}$ gives equal overdrives for all three devices. The required aspect ratios are

$$\left(\frac{W_{\text{eff}}}{L}\right)_n = \frac{2(26.581)}{215.6(0.177208)^2} = 7.8522, \quad \left(\frac{W_{\text{eff}}}{L}\right)_p = 35.5881.$$

Device	w_{ana} (μm)	L_{ana} (μm)	n_f	m	W_{eff} (μm)
M1 (NFET)	0.53002	0.27000	2	2	2.12008
M2, M3 (PFET), each	2.40220	0.27000	2	2	9.60878

The estimated gate biases are

$$V_{in,Q} = V_{SS} + V_{TN} + V_{OV1} = -0.289792\,\text{V},$$
$$V_{G2} = V_{G3} = V_{DD} - |V_{TP}| - V_{OV2} = 0.315992\,\text{V}.$$

With $V_{o,Q} = 0$, M1 and M2 each have $|V_{DS}| = 0.9\,\text{V}$; M3 is diode-connected with $V_{SD3} = 0.584008\,\text{V}$. All exceed $V_{OV} = 0.177208\,\text{V}$ under the assumed model. The mirror current will require adjustment when unequal drain voltages and DC channel-length modulation are included.

Compensation and two-node response. The Miller estimate gives $C_C \approx 0.795775\,\text{pF}$. With $f_{\text{BW}} = 2\,\text{MHz}/31.6228 = 63.2456\,\text{kHz}$, the complete transfer function satisfies the half-power condition at $\omega_B = 2\pi f_{\text{BW}}$:

$$(1 - b\omega_B^2)^2 + (a\omega_B)^2 = 2\left[1 + \left(\frac{\omega_B C_C}{G_m}\right)^2\right]. \tag{1}$$

This yields $C_m = C_C = 0.716828\,\text{pF}$ in the no-parasitic model. The resulting estimates are

$$A_{v0} = -31.6228, \quad f_{p1} = 63.2456\,\text{kHz}, \quad f_{p2} = 53.005\,\text{MHz}, \quad f_z = 66.608\,\text{MHz}.$$

The actual unity-gain crossing of this model is $f_u = 1.99848\,\text{MHz}$, where the dynamic phase lag is $92.07^\circ < 120^\circ$. Its GBW is 2.000 MHz. The external C_m is reduced by the intrinsic contribution of C_{gd1}.

(b) Digital CMOS inverter (Fig. 7-6).

Length and current selection. Both gates are driven by the input, giving $G_m = g_{mn} + g_{mp}$. The selected bias is $V_{in,Q} = V_{o,Q} = 0$, with equal lengths $L_n = L_p = L$. The overdrives are

$$V_{OVn} = 0.9 - 0.4330 = 0.4670\,\mathrm{V}, \qquad V_{OVp} = 0.9 - 0.4068 = 0.4932\,\mathrm{V}.$$

Equal drain currents require

$$\frac{\beta_p}{\beta_n} = \left(\frac{V_{OVn}}{V_{OVp}}\right)^2 = 0.89658, \qquad \beta_i = K_{Pi}\frac{W_{\mathrm{eff},i}}{L_i}.$$

The unequal thresholds mean that $\beta_n = \beta_p$ does not place the exact transition center at zero. Using the length-scaling assumption,

$$A_0 = \frac{2/V_{OVn} + 2/V_{OVp}}{\lambda_n(L) + \lambda_p(L)},$$
$$L = L_0\,\frac{A_0[\lambda_n(L_0) + \lambda_p(L_0)]}{2/V_{OVn} + 2/V_{OVp}} = 0.365476\,\mu\mathrm{m}.$$

The choice $I_D = 120\,\mu\mathrm{A}$ places the non-dominant pole and zero well above the required bandwidth in the assumed model. Then

$$g_{mn} = \frac{2(120\,\mu\mathrm{A})}{0.4670\,\mathrm{V}} = 513.919\,\mu\mathrm{S},$$
$$g_{mp} = \frac{2(120\,\mu\mathrm{A})}{0.4932\,\mathrm{V}} = 486.618\,\mu\mathrm{S},$$
$$G_m = 1000.537\,\mu\mathrm{S}, \qquad R_{\mathrm{out}} = \frac{A_0}{G_m} = 31.6058\,\mathrm{k\Omega}.$$

Device sizes and operating point. The effective width of each transistor follows from $W_{\mathrm{eff}} = 2I_D L/(K_P V_{OV}^2)$:

Device	w_{ana} (μm)	L_{ana} (μm)	n_f	m	W_{eff} (μm)
M1 (NFET)	0.46637	0.36548	2	2	1.86547
M2 (PFET)	1.89509	0.36548	2	2	7.58037

At the selected bias, $V_{DS1} = V_{SD2} = 0.9\,\mathrm{V}$, which exceeds both overdrives. The NMOS body is at V_{SS} and the PMOS body at V_{DD}; both active devices have zero source-body bias.

Compensation and performance. Solving Eq. (1) with the new G_m and R_{out} gives $C_m = C_C = 0.754672\,\mathrm{pF}$. The resulting estimates are

$$A_{v0} = -31.6228 \quad (30.000\,\mathrm{dB}),$$
$$f_{\mathrm{BW}} = 63.2456\,\mathrm{kHz}, \qquad \mathrm{GBW} = 2.000\,\mathrm{MHz},$$
$$f_{p2} = 167.913\,\mathrm{MHz}, \qquad f_z = 211.006\,\mathrm{MHz},$$
$$f_u = 1.99895\,\mathrm{MHz}, \qquad \phi_{\mathrm{lag}}(f_u) = 89.41^\circ.$$

The analytical response satisfies the phase requirement. In the physical circuit, $C_C = C_m + C_{gd1} + C_{gd2}$; gate and output capacitances also affect the AC response.

(c) PMOS-only inverter with self-biased load (Fig. 7-8).

Unity-gain sizing with finite output conductances. M1 is the upper driver. M2 has drain and gate at V_{SS}, source at V_o, and body at V_{DD}. For the initial no-body-effect calculation at $V_{o,Q} = 0$,

$$V_{OV2} = 0.9 - |V_{TP}| = 0.4932\,\mathrm{V}.$$

The handout's ideal equal-size result $V_o = -V_{in}$ neglects finite g_o. Retaining g_o gives

$$A_{v0} = -\frac{g_{m1}}{g_{m2} + g_{o1} + g_{o2}}.$$

Unity gain therefore requires $G_m = g_{m1} = G_{\mathrm{out}}$, giving $R_{\mathrm{out}} = 1/G_m$ and $A_{v0} = -1$. The simple load-dominant estimate is

$$G_m \approx 2\pi C_L \mathrm{GBW} = 314.159\,\mu\mathrm{S}.$$

For the initial choice $C_m = 0.010\,\mathrm{pF}$, Eq. (1) determines G_m at $f_{\mathrm{BW}} = 2\,\mathrm{MHz}$. For this unity-gain case,

$$a = 2R_S C_m + \frac{C_L + C_m}{G_m}, \qquad b = \frac{R_S C_L C_m}{G_m},$$

which gives $G_m = 318.387\,\mu\mathrm{S}$. The realizability of this small C_m depends on the PDK's available capacitance range and the intrinsic parasitic capacitance.

The choice $L_1 = L_2 = 0.270\,\mu\mathrm{m}$ gives $\lambda_\Sigma = 2\lambda_p = 0.3890\,\mathrm{V}^{-1}$. Since $g_{m2} = 2I_D/V_{OV2}$,

$$\begin{aligned} I_D &= \frac{G_m}{2/V_{OV2} + \lambda_\Sigma} = 71.642\,\mu\mathrm{A}, \\ V_{OV1} &= \frac{2I_D}{G_m} = 0.450030\,\mathrm{V}, \\ V_{in,Q} &= V_{DD} - |V_{TP}| - V_{OV1} = 0.043170\,\mathrm{V}. \end{aligned}$$

Applying the square-law sizing equation separately to M1 and M2:

Device	w_{ana} (μm)	L_{ana} (μm)	n_f	m	W_{eff} (μm)
M1 (PFET driver)	1.00389	0.27000	2	2	4.01555
M2 (PFET load)	0.83584	0.27000	2	2	3.34335

The driver is slightly wider to compensate the two output conductances. Here $g_{m2} = 290.519\,\mu\mathrm{S}$ and $g_{o1} + g_{o2} = 27.869\,\mu\mathrm{S}$. Both PMOS devices have $V_{SD} = 0.9\,\mathrm{V}$, exceeding their respective overdrives under the no-body-effect assumption.

The complete two-node calculation gives

$$A_{v0} = -1, \quad R_{\mathrm{out}} = 3.14083\,\mathrm{k\Omega}, \quad f_{\mathrm{BW}} = \mathrm{GBW} = 2.000\,\mathrm{MHz}.$$

Its pole frequencies are 2.00031 MHz and 161.272 MHz; the RHP zero is at 5067.29 MHz. The phase lag at f_{BW} is 45.73°. This unity-DC-gain low-pass model has no positive-frequency unity-gain crossing, so its phase is evaluated at the bandwidth.

Body-effect correction for the PMOS load. The preceding PMOS-only dimensions neglect body effect. With M2's body fixed at V_{DD} and its source at zero, $V_{BS2} = 0.9\,\mathrm{V}$. The load therefore exhibits body effect, whereas the upper driver has zero source-body bias. Let $\Phi = 2|\phi_F|$ and use the first-order relations

$$|V_{TP2}| \approx |V_{TP0}| + \gamma_p \left(\sqrt{\Phi + V_{BS2}} - \sqrt{\Phi}\right),$$
$$\eta = \frac{g_{mb2}}{g_{m2}} \approx \frac{\gamma_p}{2\sqrt{\Phi + V_{BS2}}}.$$

The corrected quantities become

$$V_{OV2} = 0.9 - |V_{TP2}|,$$
$$G_{\mathrm{out}} = g_{m2}(1+\eta) + \lambda_\Sigma I_D,$$
$$I_D = \frac{G_m}{2(1+\eta)/V_{OV2} + \lambda_\Sigma},$$
$$V_{OV1} = \frac{2}{2(1+\eta)/V_{OV2} + \lambda_\Sigma},$$
$$\frac{\beta_1}{\beta_2} = \left(1 + \eta + \frac{\lambda_\Sigma V_{OV2}}{2}\right)^2.$$

Thus M1 and M2 generally require unequal sizes even with matched PMOS process parameters. A numerical correction requires γ_p and ϕ_F, which are absent from the parameter table. The PDK operating-point values of g_m, g_{mb}, g_{ds}, and threshold voltage provide the corresponding refinement. M2's body connection remains at V_{DD} in this design.

(d) CMOS inverter with self-biased load (Fig. 7-9).

M1 is the NMOS driver. M2 is the diode-connected PMOS load, with source and body at V_{DD} and drain/gate at V_o. Both active source-body voltages are zero. At $V_{o,Q} = 0$, the load overdrive is $V_{OV2} = 0.4932\,\mathrm{V}$. The selected lengths are $L_1 = L_2 = 0.540\,\mu\mathrm{m}$, for which the assumed scaling gives

$$\lambda_\Sigma = (0.1624 + 0.1945)\frac{0.270}{0.540} = 0.17845\,\mathrm{V}^{-1}.$$

The same $C_m = 0.010\,\mathrm{pF}$ and $G_m = 318.387\,\mu\mathrm{S}$ as in part (c) apply because the selected A_0, R_S, and C_L are the same. Unity gain requires $g_{m1} = g_{m2} + g_{o1} + g_{o2}$:

$$I_D = \frac{318.387 \times 10^{-6}}{2/0.4932 + 0.17845} = 75.205\,\mu\mathrm{A},$$
$$V_{OV1} = \frac{2I_D}{G_m} = 0.472411\,\mathrm{V},$$
$$V_{in,Q} = V_{SS} + V_{TN} + V_{OV1} = 0.005411\,\mathrm{V}.$$

CMOS self-biased inverter: dimensions and response.

The continuous analytical dimensions are

Device	w_{ana} (μm)	L_{ana} (μm)	n_f	m	W_{eff} (μm)
M1 (NFET driver)	0.42201	0.54000	2	2	1.68803
M2 (PFET load)	1.75481	0.54000	2	2	7.01925

Here $g_{m2} = 304.967\,\mu\text{S}$ and $g_{o1} + g_{o2} = 13.420\,\mu\text{S}$. At $V_{o,Q} = 0$, $V_{DS1} = V_{SD2} = 0.9\,\text{V}$, so both devices are in saturation in the analytical model. The small-signal estimates are

$$\begin{aligned} A_{v0} &= -1 \quad (0\,\text{dB}), \qquad R_{\text{out}} = 3.14083\,\text{k}\Omega, \\ f_{\text{BW}} &= \text{GBW} = 2.000\,\text{MHz}, \\ f_{p1} &= 2.00031\,\text{MHz}, \qquad f_{p2} = 161.272\,\text{MHz}, \\ f_z &= 5067.29\,\text{MHz}, \qquad \phi_{\text{lag}}(f_{\text{BW}}) = 45.73^\circ. \end{aligned}$$

The calculated frequency response is the same as part (c) because the four generic-model quantities G_m, R_B, C_B, and C_C have the same selected values. Device capacitances will generally make the two responses different.

Performance with continuous analytical dimensions.

Quantity	Mirror load	Digital CMOS	PMOS load	CMOS load
I_D (μA)	26.581	120.000	71.642	75.205
$V_{in,Q}$ (V)	−0.28979	0	0.04317	0.00541
C_m (pF)	0.71683	0.75467	0.01000	0.01000
$\lvert A_{v0}\rvert$ (dB)	30.000	30.000	0.000	0.000
f_{BW} (MHz)	0.06325	0.06325	2.000	2.000
GBW (MHz)	2.000	2.000	2.000	2.000
UGF (MHz)	1.99848	1.99895	—	—
Lag at UGF (°)	92.07	89.41	—	—
Lag at f_{BW} (°)	45.12	45.04	45.73	45.73

The currents above are branch currents. The mirror-loaded inverter also draws I_{bias} in its reference branch. The summary uses unrounded dimensions and the stated approximate model, including omission of PMOS load body effect in part (c).

Grid-constrained initial dimensions.

The following initial dimensions satisfy the design grid and replace the continuous analytical values. Both w and L are selected on the 0.01 μm grid before simulation. All listed instances initially have $n_f = m = 2$, with $W_{\text{eff}} = 4w$. These initial counts remain adjustable.

Circuit	MOS	w (μm)	L (μm)	n_f	m	W_{eff} (μm)
Mirror load	M1	0.53	0.27	2	2	2.12
	M2, M3, each	2.40	0.27	2	2	9.60
Digital CMOS	M1	0.47	0.37	2	2	1.88
	M2	1.90	0.37	2	2	7.60
PMOS buffer	M1	1.00	0.27	2	2	4.00
	M2	0.84	0.27	2	2	3.36
CMOS buffer	M1	0.42	0.54	2	2	1.68
	M2	1.75	0.54	2	2	7.00

The operating points and small-signal parameters below correspond to these exact dimensions. The mirror reference current is $I_{\text{bias}} = 26.5812\,\mu$A. For the other circuits, the input bias is obtained by imposing $V_{o,Q} = 0$ on the grid-constrained dimensions. Consequently, the digital CMOS pair no longer has exactly zero input bias.

Quantity	Mirror load	Digital CMOS	PMOS buffer	CMOS buffer
I_D (μA)	26.5812	119.1389	71.9987	74.9987
$V_{in,Q}$ (V)	−0.289789	−0.000621	0.041175	0.005889
C_m (pF)	0.716827	0.754832	0.010000	0.010000
$\lvert A_{v0}\rvert$ (dB)	29.9998	30.1075	−0.0384	−0.0088
f_{BW} (MHz)	0.063247	0.062468	2.009943	1.994614
GBW (MHz)	2.000000	2.000000	2.001070	1.992597
UGF (MHz)	1.998480	1.998972	—	—
Lag at UGF (°)	92.065	89.444	—	—
Lag at f_{BW} (°)	45.123	45.039	45.732	45.727

These estimates retain the same model limitations: omitted body effect and parasitic capacitances, and an unverified $\lambda(L)$ approximation. For the grid-constrained geometry, the high-gain compensation values follow from Eq. (1) and are approximately 0.716826684 and 0.754832189 pF. The residual gain and GBW offsets follow from the selected grid-constrained dimensions. Compliance with the specifications requires simulation with the corresponding PDK devices and realizable capacitor values.

Dummy devices and physical implementation. Each active MOS has one separately named dummy instance placed beside it. Every dummy's D/G/S/B terminals connect to V_{SS}, including the PFET dummies. Dummy n_f and m remain configurable. These devices are not electrically paralleled with the active MOS, so their widths are excluded from W_{eff} in the tables. A PFET dummy requires its own V_{SS} well/tap separated from the active PFET's V_{DD} well. Each dummy retains its own integer n_f and m and legal per-finger width; the initial dummy geometry follows its associated active MOS. The physical placement must satisfy the PDK well-spacing and isolation rules.

Matching and common-centroid placement. The equal-size PMOS mirror M2/M3 uses identical units in a 2×2 AB/BA common-centroid arrangement, preserving equal centroids and consistent orientation. The unequal-width PMOS buffer uses equal-centroid interleaving of its driver and load units; this is not identical-unit matching and does not remove the devices' intentional width difference or the load's body effect. NMOS/PMOS pairs use separate well/body regions and symmetric blocks associated with each device; no cross-type common-centroid equivalence is assumed. Dummies remain adjacent to their associated active units and are not consolidated into a separate array.

Multiplicity and extracted-device correspondence. The PDK `mSwitch` convention must agree with the schematic's width, finger-count, and multiplicity interpretation. The initial schematic uses `mSwitch=1`. Its correspondence to grouped extraction for $n_f > 1$ requires CDF evaluation and comparison of schematic and extracted widths and multiplicities. Decomposition into units for common-centroid placement must preserve these same electrical quantities.

Consistency of simulated and physical dimensions. DC and AC verification use the exact grid-constrained geometry, selected PDK passive values, and intended n_f, m, and `mSwitch` settings. The schematic and layout retain these values after the specifications are met. Any subsequent change in dimensions or effective multiplicity requires renewed circuit simulation.

Model limitations and circuit verification.
The first-order model determines the initial dimensions, bias voltages, and compensation values. The PMOS-only load requires the body-effect correction above; the longer-channel CMOS designs depend on the accuracy of the assumed $\lambda(L)$ behavior. Gate capacitance, finite mirror reference impedance, junction capacitance, and physical passive models may shift both gain and bandwidth.

The DC transfer characteristic is obtained by sweeping V_{in} from -0.9 to $+0.9$ V. The two intercepts and the input bias producing $V_{out} = 0$ characterize the operating point. AC analysis about that bias gives the low-frequency gain, relative -3 dB bandwidth, gain-bandwidth product, and any positive-frequency unity-gain crossing. Dynamic phase lag excludes the DC inversion and is evaluated at UGF for the high-gain designs and at the bandwidth for the unity-gain designs. Each sizing change and parasitic extraction requires a new bias calculation.

Pre-layout simulation uses legal, grid-constrained MOS dimensions and PDK passive values. The schematic and layout share the same device parameters and connectivity, including R_S and C_m. Any dimensional change requires renewed DC and AC analysis. DRC and LVS establish geometrical legality and schematic/layout correspondence; simulation with extracted parasitic capacitances then determines the resulting circuit performance. The analytical estimates alone do not establish compliance with these requirements.

Sources.

[1] S. Palermo, *ECEN 474/704 Lab 7: Frequency Response of Inverting Amplifiers*, course laboratory handout, Figs. 7-2, 7-5, 7-6, 7-8, and 7-9; Prelab Questions 1–3.

[2] Anonymous contributor, *ECEN 474/704 Lab 1: Introduction to Cadence and MOS Device Characterization*, laboratory report, Table I: single-finger $5/0.27\ \mu$m NFET/PFET parameter measurements.